HOW WE TESTED

Our performance tests involve the use of multiple benchmarks that test read and write performance using different data signatures to simulate different types of operations. Certain benchmarks look towards a pure read or write speed for sequential files. Then we have benchmarks that simulate a normal operating system's data transfer pattern. We also make use of our own sample files which involve compressible and incompressible data of varying sizes. Our assorted sample data consists of an insane amount of files which would be akin to huge game transfers. So we not only rely on synthetic benchmarks but also real world tests, which are a better indication of what an SSD would undergo once installed in a system. Needless to say, we weigh this test quite high in our performance benchmarks. The following are the synthetic benchmarks used. We don't believe in running too many synthetic benchmarks that produce the same results.

- **AS-SSD**

For transfer speeds and IOPS in 4KQD64 thread tests.

- **Crystal Disk Mark**

Another synthetic benchmark for sequential, 512K, 4K and 4KQD32 tests.

- **PCMark 8 Storage**

An industry standard benchmark that simulates data signatures for uses like playing games, running office productivity suites, multimedia editing software and also file transfer speeds.

We list out the various features that each drive comes with and we weigh the individual parameters to gain a feature score. The overall weightage given to the feature score is meagre since SSDs are all about performance and there aren't that many differentiating features anyway.

The last thing we do is check the SSDs for build quality. This involves opening the drives and checking how the PCB is mounted within the casing and also if the controller and the NAND chips are mounted with heatsinks or thermal pads. This too is given a meagre weightage compared to the performance score.

will be way greater than what cheap controllers of today are capable of.

A NEW ENTRANT

We were surprised to find that one of the entrants in the test is an upcoming brand in India. MaStar has already released USB flash drives and memory cards and this is the first time we tested any of their products. The GL01 series uses Phison's PS3109-S9, which is an entry level controller that has been used in Corsair's LS series and also in Kingmax's SSD lineup. There was some confusion since the product documentation mentions that it has Toshiba NAND but when we popped it open, all we could find was Micron NAND. There's nothing wrong with that but a goof up in a product's documentation is rare. Maybe the GT01 has them but we can't say anything for sure till we get one for benchmarking. The GL01 was the only one with a Phison controller in this benchmark, so it was natural that it's performance would be easily eclipsed. A comparison among other Phison enabled SSDs would be a better indicator of it's performance in the entry-segment. We don't have the retail price for these SSDs yet, so haven't considered these devices in the final scoring.

THE COMPETITORS

There weren't many new competitors this time around. Kingston's Fury is their budget offering targeted towards gamers and enthusiasts. Performance wise it's average, but it's way better when it comes to build quality amongst mid-range SSDs. It uses Kingston branded NAND chips though and Kingston has no NAND manufacturing facilities. Even the controller is a Kingston branded SF2281.

Transcend's SSD370 is part of their premium lineup but performance wise we'd place it more towards the upper ceiling of the entry-level segment. Also, we were pleasantly surprised to learn that Transcend makes PATA SSDs, yes! PATA!

Seagate's 600 series was also a part of this test and it was one of the top performers in the entire test. But an MRP of ₹17,500 it's quite an expensive unit. We did, however, spot units selling for around ₹12,000 on Amazon, which brings it into the same league as the other top performers.

ADATA's SP610 turned out to be average, given the small sample size that we had to work with and it's cousin, the SP920 seemed to have lost out by a mere fraction in the performance test.

SanDisk Extreme II

WINNERS:
Best Performer
SanDisk Extreme II 240GB

There are a lot of factors contributing towards SanDisk's performance in this test. Firstly, the NAND is made by them, given the Toshiba-SanDisk partnership. The Marvell 88SS9187 is one of the best controllers which gives a consistently high performance across varying data sizes and queue depths. We saw that the SanDisk Extreme II and the Plextor M6S shared pretty much the exact same configuration and scored quite similar in a few benchmarks but the SanDisk Extreme II was almost always ahead. This can be attributed to the fact that the Marvell 88SS9188 that the Plextor M6S uses is a slightly scaled down version of the 88SS9187. ADATA's SP920SS missed out by quite a margin from winning and this was mostly because of the weightages assigned to individual benchmark scores. Our weightages are scaled towards a more consistent performance and does not favour a drive that excels in just one or two aspects. Higher write speeds is what pushed the ADATA SP920 ahead in our score.

Aside from performance, even when we opened the two drives we noted that the SanDisk Extreme II had heat conducting pads between the controller/NAND chips and the metal body. The Plextor